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[QR code to github page]

A selection of self-directed design, simulation, and software projects reflecting hands-on CAD modeling, mechanical systems thinking, and programming ability.

4-Cylinder Engine Crankshaft & Piston Assembly

[FreeCAD] | [Mechanical Design]

A complete bottom end mechanical assembly made of pistons, connecting rods, and a multi throw crankshaft, modeled to demonstrate internal combustion rotating dynamics.

The Challenge: Establishing correct kinematic constraints. If the tolerances and joints connection between the multi throw crankshaft, rods, pistons were even slightly off, the 3D model during the assembly would completely collapse and clip through each other.

The Solution: I utilized fully parametric, fully constrained 2D sketches for the base parts. By carefully applying assembly workflow, while making sure all joints were applied thoroughly, I was able to successfully translate the linear piston stroke motion into continuous rotational torque without constraint failures.

Quadcopter Drone Frame

[FreeCAD] | [UAV Design]

A complete modular, symmetrical 4 arm UAV airframe capable of safely mounting motors and an internal electronic payload.

The Challenge: Ensuring precise alignment and constraint management across multi-layer structural plates, motor mounts, and rotating hardware, and maintaining modularity so individual components (e.g., motor arms or standoffs) can be modified or replaced without breaking top-level assembly relationships.

The Solution: Used fully constrained 2D sketches and parametric dimensions in FreeCAD, carefully applied structured 3D assembly constraints to align motor shaft, propeller hubs, etcs, ensuring axial alignment.

Real time 2D Inverse Kinematics (IK) Robotic Arm

[Python] | [Pygame] | [Robotic/Kinematics]

A real-time robotic arm simulator that dynamically calculates and renders joint angles required to track complex target pathways on screen.

The Challenge: The robotic arm instantly teleporting to the new target coordinates. Furthermore, the bug with minuscule float values that caused joint angle oscillations, jitter, or full kinematics solver failures. Due to IEEE 754 precision inaccuracies, exact calculations can produce intermediate values like `cos_t2 = 1.0000000000000002`. Passing this to `acos()` immediately causes an unhandled domain exception.

The Solution: Implemented smooth step-wise motion linear interpolation (LERP) per frame to eliminate the teleportation, and placed a strict bounding clamp `[-1.0, 1.0]` on the cosine input variable prior to computing inverse trigonometric functions

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Minimal TUI/Headless Audio Player

[C++] | [low level programming] | [CMAKE]

A minimal, highly optimized, CLI and TUI media engine showcasing software engineering and technical skills. Built entirely in C++ using Miniaudio for low level audio processing and FTXUI for dynamic terminal rendering.

The Challenge: Integrating external C libraries and UI frameworks into a seamless C++ build using CMake, while handling user inputs (play, pause, skip) and external API calls without freezing the audio playback.

The Solution: Separated the terminal interface logic from the core audio playback loop using standard C++ threads and mutex locks to ensure responsive user controls. Offloaded high-latency Last.fm API network calls to a detached asynchronous process using `fork()` and `execvp()` to prevent audio stuttering